

Lecture 7

Data Definition Language (SQL DDL)

What are Constraints?

- **Constraints enforce rules at the table level.**
- **Constraints prevent the deletion of a table if there are dependencies.**
- **The following constraint types are valid:**
 - **NOT NULL**
 - **UNIQUE**
 - **PRIMARY KEY**
 - **FOREIGN KEY**
 - **CHECK**
 - **DEFAULT**

Constraint Guidelines

- Name a constraint or the Oracle server generates a name by using the `SYS_Cn` format.
- Create a constraint either:
 - At the same time as the table is created, or
 - After the table has been created
- Define a constraint at the column or table level.
- User named or system named constraint
- View a constraint in the data dictionary.

Defining Constraints

```
CREATE TABLE [schema.] table
  (column datatype [DEFAULT expr]
   [column_constraint],
   ...
   [table_constraint] [, ...]);
```

```
CREATE TABLE employees(
  employee_id  NUMBER(6),      Column level
  first_name   VARCHAR2(20),
  ...
  job_id       VARCHAR2(10) NOT NULL,
  PRIMARY KEY (EMPLOYEE_ID));
```

Table level

```
CREATE TABLE employees(
  employee_id  NUMBER(6) PRIMARY KEY,      Column level
  first_name   VARCHAR2(20),
  ...
  job_id       VARCHAR2(10) NOT NULL
);
```

Column level

Defining Constraints

- Column constraint level

```
column [CONSTRAINT constraint_name] constraint_type,
```

- Table constraint level

```
column, ...  
  [CONSTRAINT constraint_name] constraint_type  
  (column, ...),
```

The NOT NULL Constraint

Ensures that null values are not permitted for the column:

EMPLOYEE_ID	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	DEPARTMENT_ID
100	King	SKING	515.123.4567	17-JUN-87	AD_PRES	24000	90
101	Kochhar	NKOCHHAR	515.123.4568	21-SEP-89	AD_VP	17000	90
102	De Haan	LDEHAAN	515.123.4569	13-JAN-93	AD_VP	17000	90
103	Hunold	AHUNOLD	590.423.4567	03-JAN-90	IT_PROG	9000	60
104	Ernst	BERNST	590.423.4568	21-MAY-91	IT_PROG	6000	60
178	Grant	KGRANT	011.44.1644.429263	24-MAY-99	SA_REP	7000	
200	Whalen	JWHALEN	515.123.4444	17-SEP-87	AD_ASST	4400	10

...

20 rows selected.


NOT NULL constraint
(No row can contain
a null value for
this column.)


**NOT NULL
constraint**


**Absence of NOT NULL
constraint**
(Any row can contain
null for this column.)

The NOT NULL Constraint

Is defined at the column level:

```
CREATE TABLE employees(  
    employee_id    NUMBER(6) ,  
    last_name      VARCHAR2(25) NOT NULL,  
    salary         NUMBER(8,2) ,  
    commission_pct NUMBER(2,2) ,  
    hire_date      DATE  
    CONSTRAINT emp_hire_date_nn  
    NOT NULL,  
    ...
```

← System
named

← User
named

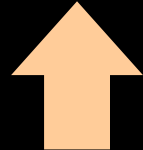
The UNIQUE Constraint

EMPLOYEES



EMPLOYEE_ID	LAST_NAME	EMAIL
100	King	SKING
101	Kochhar	NKOCHHAR
102	De Haan	LDEHAAN
103	Hunold	AHUNOLD
104	Ernst	BERNST

...



INSERT INTO

208	Smith	JSMITH
209	Smith	JSMITH



Allowed



Not allowed:
already exists

The UNIQUE Constraint

Defined at either the table level or the column level:

Table Level

```
CREATE TABLE employees(  
    employee_id    NUMBER(6),  
    last_name      VARCHAR2(25) NOT NULL,  
    email          VARCHAR2(25),  
    salary         NUMBER(8,2),  
    commission_pct NUMBER(2,2),  
    hire_date      DATE NOT NULL,  
    CONSTRAINT emp_email_uk UNIQUE(email));
```

```
CREATE TABLE employees(  
    employee_id    NUMBER(6),  
    last_name      VARCHAR2(25) NOT NULL,  
    email          VARCHAR2(25),  
    salary         NUMBER(8,2),  
    commission_pct NUMBER(2,2),  
    hire_date      DATE NOT NULL,  
    UNIQUE(email));
```

Column Level

```
CREATE TABLE employees(  
    employee_id    NUMBER(6),  
    last_name      VARCHAR2(25) NOT NULL,  
    email          VARCHAR2(25)  
        CONSTRAINT emp_email_uk  
        UNIQUE,  
    salary         NUMBER(8,2),  
    commission_pct NUMBER(2,2),  
    hire_date      DATE NOT NULL,  
    );
```

```
CREATE TABLE employees(  
    employee_id    NUMBER(6),  
    last_name      VARCHAR2(25) NOT NULL,  
    email          VARCHAR2(25) UNIQUE,  
    salary         NUMBER(8,2),  
    commission_pct NUMBER(2,2),  
    hire_date      DATE NOT NULL,  
    );
```

The PRIMARY KEY Constraint

DEPARTMENTS

 **PRIMARY KEY**

DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
10	Administration	200	1700
20	Marketing	201	1800
50	Shipping	124	1500
60	IT	103	1400
80	Sales	149	2500

...

**Not allowed
(Null value)**



INSERT INTO

	Public Accounting		1400
50	Finance	124	1500

**Not allowed
(50 already exists)**



The PRIMARY KEY Constraint

Defined at either the table level or the column level:

Table Level

```
CREATE TABLE departments(  
  department_id      NUMBER(4) ,  
  department_name    VARCHAR2(30 ,  
  manager_id        NUMBER(6) ,  
  location_id        NUMBER(4) ,  
  CONSTRAINT dept_name_nn NOT NULL,  
  CONSTRAINT dept_id_pk  
    PRIMARY KEY(department_id)  
);
```

Column Level

```
CREATE TABLE departments(  
  department_id NUMBER(4)  
    CONSTRAINT dept_id_pk PRIMARY KEY,  
  department_name VARCHAR2(30)  
    CONSTRAINT dept_name_nn NOT NULL,  
  manager_id      NUMBER(6) ,  
  location_id     NUMBER(4) ,  
);
```

```
CREATE TABLE departments(  
  department_id      NUMBER(4) ,  
  department_name    VARCHAR2(30) ,  
  manager_id        NUMBER(6) ,  
  location_id        NUMBER(4) ,  
  NOT NULL (department_name) ,  
  PRIMARY KEY(department_id)  
);
```

```
CREATE TABLE departments(  
  department_id NUMBER(4) PRIMARY KEY ,  
  department_name VARCHAR2(30) NOT NULL ,  
  manager_id      NUMBER(6) ,  
  location_id     NUMBER(4) ,  
);
```

The PRIMARY KEY Constraint

- Primary Key with Composite Key

```
CREATE TABLE project_assignment(  
    project_id REFERENCE project(project_id),  
    staff_id REFERENCE staff(staff_id),  
    date_start DATE,  
    date_end DATE,  
    duration INTEGER(3),  
    PRIMARY KEY(project_id,staff_id)  
);
```

The FOREIGN KEY Constraint

DEPARTMENTS

DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
10	Administration	200	1700
20	Marketing	201	1800
50	Shipping	124	1500
60	IT	103	1400
80	Sales	149	2500

**PRIMARY
KEY**



...



EMPLOYEES

EMPLOYEE_ID	LAST_NAME	DEPARTMENT_ID
100	King	90
101	Kochhar	90
102	De Haan	90
103	Hunold	60
104	Ernst	60
107	Lorentz	60

**FOREIGN
KEY**



...

INSERT INTO



200	Ford	9
201	Ford	60

**Not allowed
(9 does not
exist)**

Allowed

ORACLE

The FOREIGN KEY Constraint

Defined at either the table level or the column level:

```
CREATE TABLE employees(  
    employee_id      NUMBER(6) ,  
    last_name        VARCHAR2(25) NOT NULL,  
    email            VARCHAR2(25) ,  
    salary           NUMBER(8,2) ,  
    commission_pct   NUMBER(2,2) ,  
    hire_date        DATE NOT NULL,  
    ...  
    department_id    NUMBER(4) ,  
    CONSTRAINT emp_dept_fk FOREIGN KEY (department_id)  
        REFERENCES departments(department_id) ,  
    CONSTRAINT emp_email_uk UNIQUE(email)) ;
```

The CHECK Constraint

- Defines a condition that each row must satisfy

```
..., salary    NUMBER(2)
      CONSTRAINT emp_salary_min
      CHECK (salary > 0),...
```

```
CREATE TABLE employees(
  employee_id      NUMBER(6),
  last_name        VARCHAR2(25) NOT NULL,
  email            VARCHAR2(25),
  salary           NUMBER(8,2),
  commission_pct   NUMBER(2,2),
  hire_date        DATE NOT NULL,
  department_id    NUMBER(4),
  CONSTRAINT emp_dept_fk FOREIGN KEY (department_id)
    REFERENCES departments(department_id),
  CONSTRAINT emp_email_uk UNIQUE(email),
  CONSTRAINT emp_salary_chk CHECK(salary>1200)
);
```

Adding a Constraint Syntax

Use the **ALTER TABLE** statement to:

- Add or drop a constraint, but not modify its structure
- Enable or disable constraints
- Add a **NOT NULL** constraint by using the **MODIFY** clause

```
ALTER TABLE table  
ADD [CONSTRAINT constraint] type (column);
```


Adding a Constraint

Add a FOREIGN KEY constraint to the EMPLOYEES table indicating that a manager must already exist as a valid employee in the EMPLOYEES table.

```
ALTER TABLE      employees
ADD CONSTRAINT    emp_manager_fk
    FOREIGN KEY (manager_id)
    REFERENCES employees (employee_id) ;
Table altered.
```

Dropping a Constraint

- Remove the manager constraint from the EMPLOYEES table.

```
ALTER TABLE      employees
DROP CONSTRAINT    emp_manager_fk;
Table altered.
```